Mark scheme — Papers A to D

Before you start

- Each paper is out of **30**. Medium papers are 40 minutes, hard papers 50 minutes, all non-calculator.
- Award method marks for a correct approach with an arithmetic slip, and follow through errors into later parts.
- Where a question asks for a percentage **change**, the words *increase* or *decrease* are part of the accuracy mark.
- Where a question says **simplest form** or **mixed number**, an unsimplified correct value scores the method mark only.

PAPER A — MEDIUM

1	**(a) 0.75 (b) 0.07 (c) 0.4**	One mark each. (b) is the one they drop — 7% is 7 hundredths, so 0.07, not 0.7.
2	**35% and 7/20**	M1 for 35/100; A1 for 7/20 fully simplified. 35/100 alone scores 1 of 2.
3	**5/6**	M1 common denominator 6 (3/6 + 2/6); A1 for 5/6.
4	**70**	10% = 35, so 20% = 70. Accept 350 × 0.2.
5	**3/4**	Divide both by 6. 9/12 scores 0 — the question said simplest form.
6	**0.7, 72%, 3/4**	M1 for converting to 0.7, 0.72, 0.75; A1 for the correct order written in the original forms.
7	**2/15**	6/45 simplified. Accept 6/45 only if simplification follows.
8	**(a) ₹180 (b) ₹1680**	(a) 10% = 150 and 1% = 15, so 12% = 150 + 30 = 180. (b) 1500 + 180 = 1680, or 1500 × 1.12 directly.
9	**(a) 15 (b) 62.5%**	(a) 40 ÷ 8 = 5, × 3 = 15. (b) 25 do not wear glasses; M1 for 25/40, A1 for 62.5%.
10	**7/12**	M1 for 10/12 − 3/12; A1 for 7/12.
11	**660**	M1 for 25% = 220 or × 0.75; A1 for 660.
12	**Maths (85% against 84%)**	M1 for one correct percentage; A1 for both; A1 for the conclusion. A conclusion with no percentages scores 0.
13	**0.625 and 62.5%**	One mark each.
14	**1000 ml**	M1 for 2/5 left, or 1.5 litres used; A1 for 1000 ml. Answer in litres only loses the second mark.
15	**12.5% increase**	M1 for 80/640; A1 for 12.5%. Dividing by 720 gives 11.1% — no marks.

PAPER B — MEDIUM

1	**(a) 8% (b) 25% (c) 35%**	One mark each. (a) trips students who write 0.08%.
2	**13/20**	M1 for 65/100; A1 for 13/20.
3	**5/6**	M1 for 4/6 + 1/6; A1 for 5/6.

4	**39**	M1 for 10% = 26 and 5% = 13; A1 for 39. Accept 260 × 0.15.
5	**2/5, 43%, 0.45**	M1 for 0.4, 0.43, 0.45; A1 for the order in the original forms.
6	**1 1/5**	M1 for 4/5 × 3/2; A1 for 12/10 = 6/5 = 1 1/5.
7	**324**	M1 for × 1.35 or 35% = 84; A1 for 324.
8	**144**	M1 for × 0.8 or 20% = 36; A1 for 144.
9	**(a) 420 (b) 180**	(a) M1 for 270 and 150; A1 for 420. (b) 600 − 420 = 180.
10	**3/8**	M1 for 375/1000; A1 for 3/8.
11	**20% decrease**	M1 for 50/250; A1 for 20%. The word 'decrease' is expected.
12	**1 3/4**	M1 for 7/5 × 5/4; M1 for 35/20; A1 for 1 3/4. Multiplying the whole numbers separately scores 0.
13	**1250 g**	M1 for 3/4 ÷ 12 = 1/16 kg per biscuit, or × 20/12; A1 for 1.25 kg = 1250 g. Answer in kg loses a mark.
14	**5/8, because 5/8 = 0.625 > 0.6**	M1 for the conversion; A1 for the conclusion. A bare 'yes' scores 0.

PAPER C — HARD

1	**0.41666... = 0.41 with the 6 recurring**	M1 for a correct division to at least three decimal places; A1 for correct recurring notation (dot over the 6).
2	**7/12**	M1 for 7/3 − 7/4; M1 for 28/12 − 21/12; A1 for 7/12. Two marks available — award M1 A1.
3	**2450**	M1 for 1960 ÷ 0.8; A1 for 2450. Adding 20% to 1960 gives 2352 and scores 0.
4	**1.25 × 0.8 = 1.00, so the price is unchanged**	M1 for both multipliers; M1 for multiplying them; A1 for the conclusion, or full numerical working (e.g. 100 → 125 → 100).
5	**(a) 70 (b) 17.5**	(a) M1 for 42 ÷ 3 = 14 (one fifth); A1 for 70. (b) 70 ÷ 4 = 17.5.
6	**(a) P = 2160, Q = 2160 (b) they cost the same (c) 40%**	(a) M1 each. (b) The equal answer is deliberate — a candidate who assumes a difference and states one loses this mark. (c) 2/3 × 0.9 = 0.6, so 40% off.
7	**(a) the 10% is taken off a larger number than it was added to (b) 79.2**	(a) Must refer to the two percentages being of different amounts. (b) M1 for 1.1 × 0.9 = 0.99 or 88 − 8.8; A1 for 79.2.
8	**65.5%, then 0.66 = 33/50 equal, then 2/3**	M1 for 0.655, 0.66, 0.66, 0.6667; A1 for the order; A1 for explicitly stating that 0.66 and 33/50 are equal. Missing the equality caps this at 2.
9	**(a) 22% (b) 585**	(a) M1 for profit = 99; A1 for 99/450 × 100 = 22%. (b) M1 for × 1.3; A1 for 585.
10	**(a) 1/4 (b) 120 (c) 25%**	(a) M1 for 5/8 × 2/5; A1 for 1/4 (accept 10/40). (b) 480 ÷ 4 = 120. (c) 25% — follow through from (a).

PAPER D — HARD

1	**2 1/6**	M1 for 13/4 ÷ 3/2; M1 for 13/4 × 2/3 = 26/12; A1 for 2 1/6.
2	**840**	M1 for 966 ÷ 1.15; A1 for 840. Check: 840 × 1.15 = 966.
3	**2000**	M1 for 2400 ÷ 1.2; A1 for 2000. Taking 20% off 2400 gives 1920 and scores 0.
4	**0.175 and 17.5%; 40 = 2 × 2 × 2 × 5, so its only prime factors are 2 and 5**	One mark for the decimal, one for the percentage, one for a reason referring to the prime factors of the denominator being only 2s and 5s.
5	**(a) 10 200 (b) 12 240 (c) 2% increase**	(a) × 0.85. (b) × 1.2. (c) M1 for 0.85 × 1.2 = 1.02 or comparing 12 240 with 12 000; A1 for '2% increase' — the word 'increase' is required.
6	**(a) 75%, 72%, 70% (b) Meera, Ravi, Sana (c) 5 percentage points**	(a) M1 for 45/60 = 75%; A1 for all three. (b) One mark for the ranking. (c) 75 − 70 = 5.
7	**100 litres**	M1 for 7/10 − 2/5 = 3/10; M1 for 3/10 of capacity = 30; M1 for 30 ÷ 3 = 10 being one tenth; A1 for 100 litres.
8	**(a) 1.2 (b) 20% profit**	(a) M1 for 1.6 and 0.75; A1 for 1.6 × 0.75 = 1.2. (b) M1 for recognising 1.2 means 120% of cost; A1 for 20% profit. Answering '35% profit' (60 − 25) scores 0.
9	**(a) 4/15 (b) 900**	(a) M1 for 2/3 left after books; A1 for 40% of 2/3 = 4/15. (b) M1 for the remaining fraction being 0.6 × 2/3 = 2/5; A1 for 360 ÷ 2/5 = 900.